BME303- MATERIALS ENGINEERING

CO-1: Students will be able to identify the crystal structure and measure the mechanical properties of materials. **K3**
CO-2: Students will be able to test the various failures of materials. **K3**
CO-3: Students will be able to identify the mechanical properties based on composition of micro-constituents depicted in the phase-diagram. **K3**
CO-4: Students will understand the concept of improving the mechanical properties through heat treatment. **K2**
CO-5: Students will learn the structure and properties of alloys and composites. **K2**

UNIT-I (08 Lecture)

Crystal Structure: Unit cells, Metallic crystal structures, Ceramics. Imperfection in solids: Point, line, interfacial and volume defects; dislocation strengthening mechanisms and slip systems, critically resolved shear stress. Mechanical Properties and its measurement: Tensile, compression and torsion tests; Young's modulus, relations between true and engineering stress-strain curves, generalized Hooke's law, yielding and yield strength, ductility, resilience, toughness and elastic recovery; Hardness: Rockwell, Brinell and Vickers and their relation to strength.

UNIT-II (08 Lecture)

Alloys, substitutional and interstitial solid solutions- Hume-Rothery Rule, Phase diagrams: Interpretation of binary phase diagrams and micro-structure development; Lever Rule, eutectic, eutectoid, peritectic, peritectoid and monotectic reactions. Iron and Iron-carbide phase diagram, microstructural aspects of ledeburite, austenite, ferrite and cementite, cast iron, High Speed Steel

UNIT-III (08 Lecture)

Alloying of steel, properties of stainless steel and tool steels, maraging steels- cast irons; grey, white, malleable and spheroidal cast irons-copper and copper alloys; brass, bronze and cupro-nickel; Aluminium and Al-Cu – Mg alloys- Nickel based superalloys and Titanium alloys, Carbon Fibre and alloys, Structure and properties of Graphene Material, Composite Materials, Reinforcement

UNIT-IV (08 Lecture)

Heat treatment of Steel: Annealing, tempering, normalising and spheroidising, isothermal transformation diagrams for Fe-C alloys and microstructure development. Continuous cooling curves and interpretation of final microstructures and properties austempering, martempering, case hardening, carburizing, nitriding, cyaniding, carbo-nitriding, flame and induction hardening, vacuum and plasma hardening.

UNIT-V (08 Lecture)

Creep and failures, Introduction to non-destructive testing (NDT): Visual Inspection, Liquid Penetrate Testing, Magnetic Particle Inspection, Ultra-sonic Testing, Acoustic Testing, Radiography, X-ray Differaction, Eddy Current Testing etc., Material behaviour / properties at a very low (Cryogenics) and high temperature.

Books and References:

1. W. D. Callister, 2006, "Materials Science and Engineering-An Introduction", 6th Edition, Wiley
2. Kenneth G. Budinski and Michael K. Budinski, "Engineering Materials", Prentice Hall of India Private Limited, 4th Indian Reprint, 2002.
3. V. Raghavan, "Material Science and Engineering', Prentice Hall of India Private Limited, 1999.
4. Mechanics of materials by James M. Gere.
5. Introduction to engineering materials by B.K. Agarwal.
6. Physical metallurgy and advanced materials by R.E. Smallman.
7. Engineering mechanics of composite materials by Isaac M. Daniel.
8. U. C. Jindal, "Engineering Materials and Metallurgy", Pearson, 2011.
9. Introduction to Materials Science For Engineers by Shackelford, 9th edn, 2023, Pearson Education
10. Engineering Materials: Properties and Selection, 9edn by Budinski, Pearson Education

https://www.notebookpresentation.com/